

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Paper 2 Structured and Free Response Questions

14 September 2011

2 hours

Section A: Structured Questions

Candidates answer Section A on the Question Paper

Section B: Free Response Questions

Additional Materials: Writing Papers
Graph Paper
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group on all the work you hand in.

Write in dark blue or black pen.

You may use pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A: Structured Questions (40m)

Answer **all** questions in the space provided.

Section B: Free Response Questions (40m)

Answer **two** questions on separate writing papers.

You are advised to show all working in calculations.

You are reminded of the need for good English and clear presentation in your answers.

You are reminded of the need for good handwriting.

Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets []
at the end of each question or part question.

At the end of the examination, fasten all your work
securely together.

For Examiner's Use	
Section A	
1	10
2	9
3	9
4	12
Significant Figures	
Handwriting	
Total	40

This document consists of **15** printed pages and **1** blank page.



Section A

Answer **ALL** questions in this section on the space provided.

- 1 (a) This part of the question is about the chemistry of Period 3 elements and their compounds.

- (i) Describe what you see when magnesium burns in oxygen and write the equation for the reaction described.

.....

- (ii) Describe what you see when magnesium oxide reacts with water containing Universal Indicator. Write the equation for the reaction described.

.....

- (iii) Account for the difference in structure and bonding between magnesium and silicon.

.....

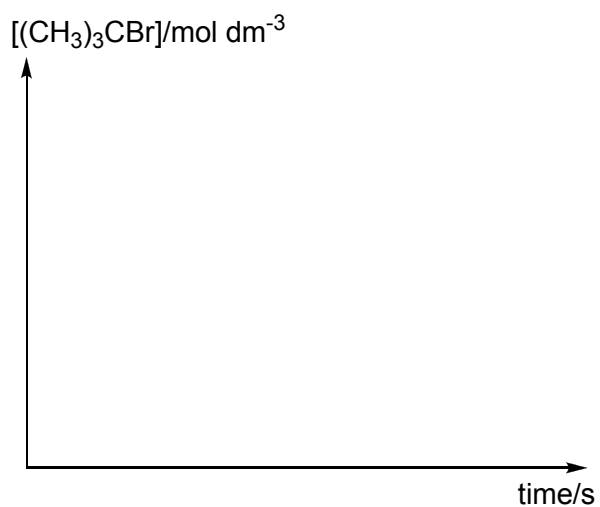
[5]

- (b) The rate of reaction between 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, and aqueous sodium hydroxide when heated under reflux is found to be independent of $[\text{OH}^-]$ but directly proportional to $[(\text{CH}_3)_3\text{CBr}]$. Hence the rate equation is

$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$$

- (i) Given that the rate constant for the reaction is $1.09 \times 10^{-3} \text{ s}^{-1}$, calculate the time taken for $(\text{CH}_3)_3\text{CBr}$ to decrease to 6.25% of its original concentration.

- (ii) Sketch the graph of concentration of $(\text{CH}_3)_3\text{CBr}$ versus time.



- (iii) With the aid of an appropriate sketch of the Maxwell-Boltzmann distribution curve, explain why an increase in temperature can speed up this chemical reaction.

.....

.....

.....

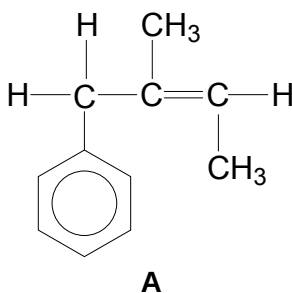
.....

.....

[5]

[Total: 10]

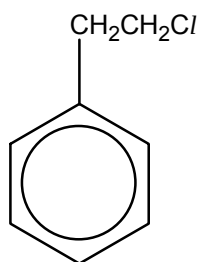
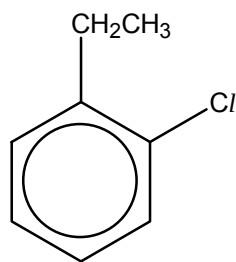
- 2 (a) Draw the structures of the major organic product(s) formed when **A** reacts with each of the following reagents.



Reagents and conditions	Major organic product(s)
(i) HBr(g)	
(ii) Alkaline KMnO ₄ , reflux	
(iii) Cl ₂ (g)/AlCl ₃ , in the dark	

[5]

- (b) Describe a simple chemical test that can be used to distinguish between **B** and **C**. State any observation(s) made and write equation(s) where appropriate.

**B****C**

Reagent(s) and Condition(s):

.....

.....

Observation for **B**:

Observation for **C**:

Equations:

.....

[4]

[Total: 9]

3 (a) The pH of a $0.100 \text{ mol dm}^{-3}$ solution of chloroethanoic acid is 2.49.

(i) Calculate the concentration of hydrogen ions in the solution.

(ii) Explain whether chloroethanoic acid is a strong or weak acid.

.....

.....

.....

(iii) The K_a expression of chloroethanoic acid is as follows:

$$K_a = \frac{[\text{CH}_2\text{C}/\text{COO}^-][\text{H}^+]}{[\text{CH}_2\text{C}/\text{COOH}]}$$

Use the K_a expression given to calculate the value of K_a for chloroethanoic acid.

[4]

(b) In an experiment, 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ solution of chloroethanoic acid was completely neutralised by $0.100 \text{ mol dm}^{-3}$ of aqueous sodium hydroxide.

(i) Calculate the number of moles and volume of sodium hydroxide needed for neutralisation.

- (ii) Suggest the indicator required and the appropriate colour change for this titration.

.....
[3]

- (c) State and explain the acid strength of chloroethanoic acid compared with that of ethanoic acid.

.....
.....
.....
.....
[2]

[Total: 9]

- 4 (a) A college student carried out an experiment to determine the standard enthalpy change of combustion of methylacetylene, $\text{CH}_3\text{C}\equiv\text{CH}$ contained in the gas cylinder of a gas stove.

A large beaker of water was placed on the stove and heated. The initial and final temperatures were recorded. The cylinder was weighed before and after the experiment to determine the mass of methylacetylene used. This process was known to be only 70% efficient. The following data were recorded by the student.

Mass of methylacetylene before heating = 4.66 g

Mass of methylacetylene after heating = 2.16 g

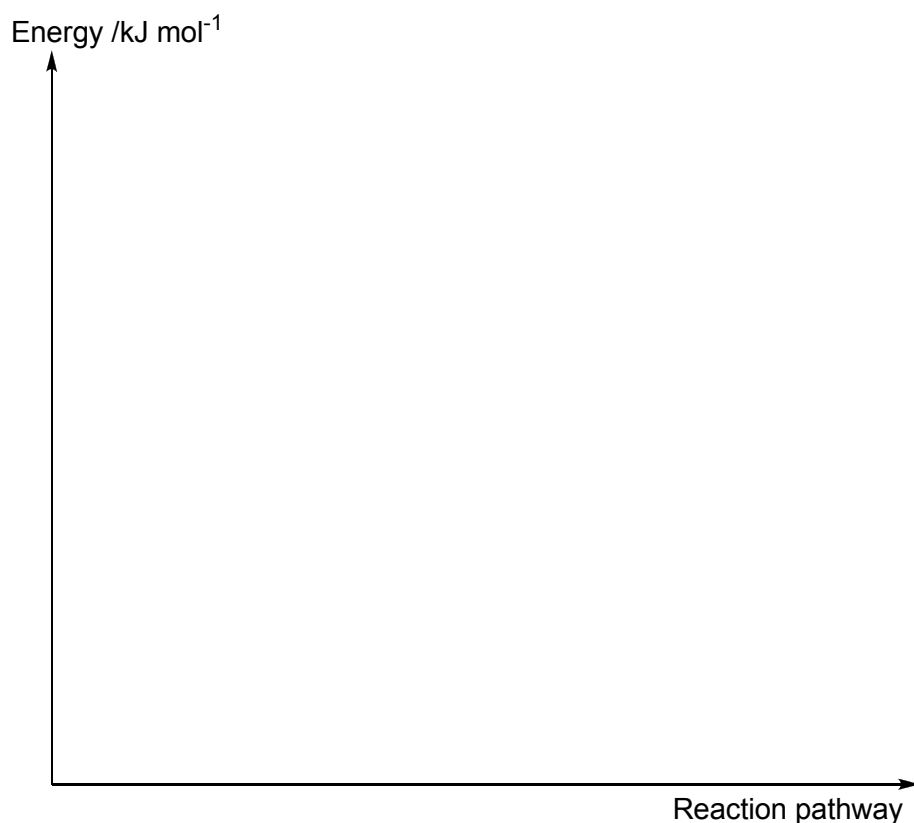
Mass of water heated = 750 g

Initial Temperature = 23.0°C

Final Temperature = 48.8°C

- (i) Calculate the heat absorbed by the water.
- (ii) Calculate the heat given out by methylacetylene during the experiment.
- (iii) Calculate the enthalpy change of combustion of methylacetylene.

- (iv) Use the information above to construct the reaction pathway diagram for the combustion of methylacetylene. Label the E_a and ΔH clearly.



[5]

- (b) Potassium iodide dissolves in water while iodine does not. However, iodine crystals dissolve readily in a solution containing potassium iodide due to its reaction with iodide ions to form I_3^- ions.

- (i) Draw the dot-and-cross diagram for the ion, I_3^- .

- (ii) Although the I_3^- ion exists, the F_3^- ion does not. Explain why the F_3^- ion does not exist.

.....

[2]

- (c) A 0.315 g sample of a Group II carbonate, ECO_3 was reacted with 25.0 cm³ of 0.108 mol dm⁻³ H_2SO_4 . The solution was boiled off to remove all the dissolved CO_2 and was then titrated with 0.124 mol dm⁻³ NaOH . It was found that 17.75 cm³ of the aqueous NaOH was required for complete neutralisation with excess H_2SO_4 .

- (i) Write an equation for the reaction taking place between the Group II carbonate, ECO_3 and aqueous sulfuric acid.

.....

- (ii) Write an equation for the reaction taking place between NaOH and H_2SO_4 during titration.

.....

- (iii) Calculate the number of moles of sulfuric acid that has reacted with sodium hydroxide during titration.

- (iv) Calculate the number of moles of sulfuric acid reacted with the metal carbonate, ECO_3 .

- (v) Calculate the M_r of the metal carbonate, ECO_3 .

[5]

[Total: 12]

Section B

Answer **two** of the three questions in this section on separate writing papers.

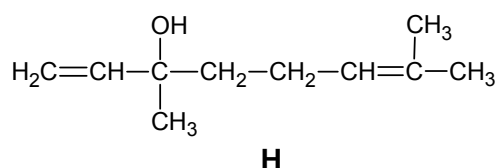
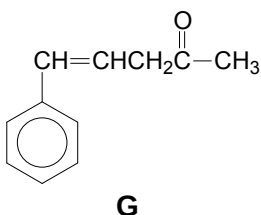
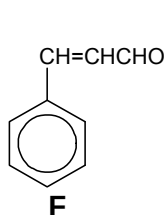
Begin each answer on a fresh sheet of paper.

- 5 (a) Ethanoic acid can be synthesised from bromomethane. Outline a 2-step synthesis to obtain ethanoic acid from bromomethane, stating clearly the reagents and conditions used in each synthesis. Identify the intermediate.

[3]

- (b) Recent research has suggested that cinnamon could be an effective pesticide against the larvae of mosquitoes, thus helping in the fight against malaria.

Each of the following three compounds, which are present in cinnamon, appears to be effective as a pesticide.



For each of the compounds **F** to **H**, state the reagents and conditions which would distinguish it from the other two and describe the observations that would be seen.

[6]

- (c) When compound **J** with a molecular formula of $C_{10}H_{12}$ is treated with cold dilute alkaline potassium manganate(VII), **J** forms compound **K**.

However, on heating **J** with acidified potassium manganate(VII), 2 products, compound **L**, C_8H_8O and compound **M** are formed. **M** has the following composition by mass: C, 40.0%; H, 6.7%; O, 53.3%.

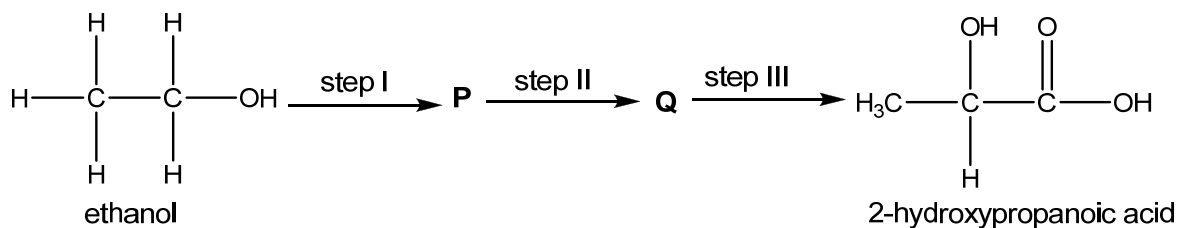
On refluxing **K** with ethanoic acid in the presence of concentrated sulfuric acid, compound **N**, $C_{14}H_{18}O_4$ is obtained.

- (i) Calculate the empirical formula of compound **M** and hence determine its molecular formula, given that the M_r of compound **M** is 60.0.
- (ii) Draw the structures of **K**, **L** and **M** and explain all the reactions above.
- (d) Explain, in terms of structure and bonding, why the melting points of Na_2O and MgO are higher than $900^\circ C$ but the melting point of P_4O_6 is less than $500^\circ C$.

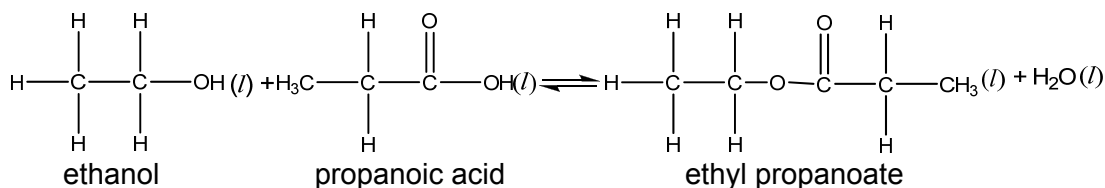
[3]

[Total:20]

- 6 The reaction scheme below shows the synthesis of 2-hydroxypropanoic acid from ethanol.



- (a) (i) State the reagents and conditions used for step I, II and III.
- (ii) Give the structures of **P** and **Q**.
- (iii) Describe a simple chemical test to distinguish between propanoic acid and 2-hydroxypropanoic acid.
- (iv) 2-hydroxypropanoic acid dimerises in an organic solvent such as tetrachloromethane. Draw a clearly labelled diagram of the dimer.
- [9]
- (b) Equal amount of ethanol and propanoic acid were mixed and allowed to reach equilibrium at temperature T. One-third of the acid was found to be in the equilibrium mixture.



- (i) Calculate the equilibrium constant, K_c , for the reaction at temperature T.
- (ii) How will the yield of ester formed be affected if anhydrous calcium chloride is added to the mixture?
- (iii) Sketch a rate-against-time graph for the forward and reverse reactions from t_0 , time when the reagents are mixed, to t_{eqm} , time when the reaction reaches equilibrium. Label the two lines clearly.
- [7]
- (c) In order to study the amount of acid left in the equilibrium mixture, the equilibrium mixture from (b) was titrated rapidly with $0.5 \text{ mol dm}^{-3} \text{ NaOH}$. Given that 25.0 cm^3 of the equilibrium mixture required 31.50 cm^3 of NaOH for complete neutralisation, what is the original concentration of propanoic acid amount in a 25.0 cm^3 sample?
- [2]

- (d) Some data for eight consecutive elements, represented by **R** to **Y** are given below.

Element	R	S	T	U	V	W	X	Y
Atomic Number	n	n+1	n+2	n+3	n+4	n+5	n+6	n+7
1 st Ionisation energy/kJ mol ⁻¹	1090	1400	1310	1680	2080	494	736	577

Given that the elements do not belong to the same period and using the information above, state the Group numbers of element **V** and **W**.

[2]

[Total:20]

- 7 Chromates have been traditionally used in chrome plating to protect metals from corrosion and to improve paint adhesion. However, due to toxicity issues, its uses have become more limited. One such compound is sodium chromate(VI), Na_2CrO_4 , a green solid.

When mixed with dilute H_2SO_4 , it disproportionates to yield $\text{Cr}^{3+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$. The mixture produced was then mixed with excess KI. The I_2 liberated required 40.0cm^3 of $0.25\text{ mol dm}^{-3} \text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ for reaction.

The 3 reactions described are as follows:

Reaction 1: Disproportionation of $\text{CrO}_4^{3-}(\text{aq})$ to yield $\text{Cr}^{3+}(\text{aq})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$

Reaction 2: $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{I}^-(\text{aq}) + 14\text{H}^+(\text{aq}) \longrightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{I}_2(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$

Reaction 3: $\text{I}_2(\text{aq}) + 2\text{S}_2\text{O}_3^{2-}(\text{aq}) \longrightarrow 2\text{I}^-(\text{aq}) + \text{S}_4\text{O}_6^{2-}(\text{aq})$

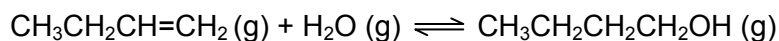
- (a) (i) Write the 2 balanced half equations occurring in **Reaction 1** in an acidic medium:
 I) CrO_4^{3-} is reduced to Cr^{3+}
 II) CrO_4^{3-} is oxidised to $\text{Cr}_2\text{O}_7^{2-}$
- (ii) Use the half-equations you have written in **a(i)** to write the overall equation with state symbols for **Reaction 1**.
- (iii) You are given that 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ liberates iodine which requires 6 moles of $\text{S}_2\text{O}_3^{2-}$ for reaction. Calculate the amount of $\text{Cr}_2\text{O}_7^{2-}$ that is formed from the disproportionation reaction.
- (iv) Hence, calculate the mass of Na_2CrO_4 used in this reaction.

[7]

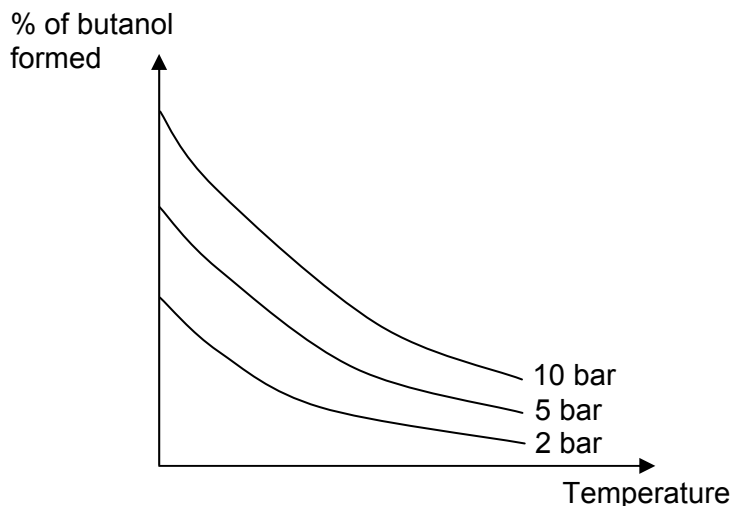
- (b) Besides transition elements, the main group elements and their oxides are also well-studied.
- (i) Sketch the general trend of the first ionisation energy of the elements across the third period of the Periodic Table. Explain the general shape of your graph.
- (ii) When the chlorides of Mg and P are dissolved in water, the pH of the resulting solutions are 6.5 and 2 respectively. Explain these reactions and write equations with state symbols where appropriate.

[6]

- (c) The hydration of gaseous butene to gaseous butanol is represented by the following equilibrium.



The percentage of butanol formed at equilibrium at various temperatures and pressures is shown below.

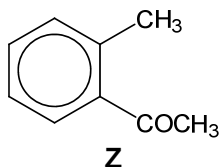


With reference to the graph above, explain the trend in the percentage of butanol formed with changes in

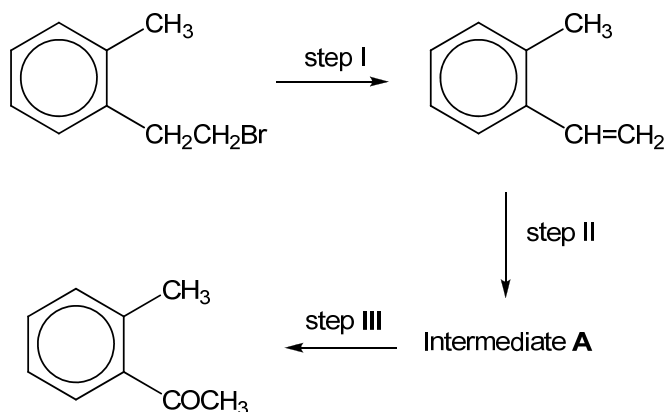
- (i) Temperature
(ii) Pressure

[3]

- (d) Compound **Z** has the following structure:



The following synthetic route shows how compound **Z** can be obtained in a three-step synthesis. State the reagents and conditions for **step I to III** and draw the structure of intermediate **A**.



[4]

